# Math 4261: Mathematical Statistics I

Lecture 16: Bayesian inference II
(References: Rice 8.5–8.6; DS 7.2–7.4)

Fall 2026

# Outline

Conjugate priors

Point estimates

More posterior examples

Bayes credible sets

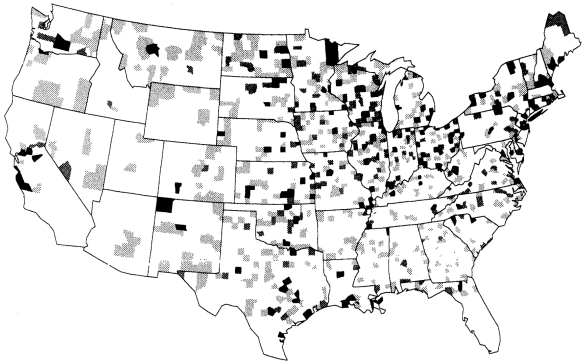
# Recap: Kidney cancer in US

An issue of sample sizes.

- Kidney cancer is rare
- Great Plains has many low-population counties, so overrepresented in the unadjusted maps.

Bayesian methods provide principled ways to:

- Adjust for county size and rarities in our estimates
- Incorporate *prior* information (e.g. national cancer rates) in our estimates.



Adjusted rates. BDA3–Gelman et al. and Manton et al.(1989)

# Recap

Given data $\boldsymbol{X} \sim f(\boldsymbol{x} \mid \theta)$ from a parametric model, Bayesian inference models $\theta$ as random, with a prior distribution

$$\Theta \sim f_{\Theta}(\theta)$$

Bayesian inference centers around the posterior distribution

$$f_{\Theta|\boldsymbol{X}}(\theta \mid \boldsymbol{x})$$

representing our updated beliefs about $\Theta$, after ('post') observing the data $\boldsymbol{X} = \boldsymbol{x}$.

The posterior may be computed from

$$f_{\Theta|\boldsymbol{X}}(\theta \mid \boldsymbol{x}) \propto f_{\boldsymbol{X}|\Theta}(\boldsymbol{x} \mid \theta) \cdot f_{\Theta}(\theta)$$

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

# Recap: Proportionality determines the distribution

It is the *shape* of a distribution that determines what it is. Scaling the probabilities by a constant does not change the shape.

For example, suppose a PDF satisfies

$$f(x) \propto \exp(-x^2/2), \qquad x \in \mathbb{R}.$$

This means $f(x) = c\exp(-x^2/2)$, where $c$ does not depend on $x$. The constant is determined by

$$1 = \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = c\sqrt{2\pi}.$$

Thus $f(x) = \dfrac{1}{\sqrt{2\pi}}\exp(-x^2/2)$: the standard Normal PDF.

In computing a posterior, we may omit factors that do not depend on the parameter, then aim to recognize the distribution.

# Reference distributions

$Y \sim \text{Beta}(\alpha, \beta)$ if $Y$ is continuous with PDF

$$f_Y(y) = \frac{1}{B(\alpha, \beta)} y^{\alpha-1}(1-y)^{\beta-1}, \quad \text{for } y \in (0,1)$$

where $B$ is the Beta function. $\mathbb{E}[Y] = \frac{\alpha}{\alpha+\beta}$.

$Y \sim \text{Gamma}(\alpha, \beta)$ if $Y$ is continuous with PDF

$$f_Y(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}, \quad \text{for } y > 0$$

where $\Gamma$ is the Gamma function. $\mathbb{E}[Y] = \frac{\alpha}{\beta}$.

# Recap: Beta–Bernoulli model

Let $X_1, \ldots, X_n \sim \mathsf{Bern}(p)$ i.i.d. with prior $P \sim \mathrm{Beta}(\alpha, \beta)$. We found that the posterior distribution is

$$P \mid \boldsymbol{X} \sim \mathrm{Beta}(S + \alpha, n - S + \beta),$$

where $S = X_1 + \cdots + X_n$.

# Example: Beta-Bernoulli–Placenta previa

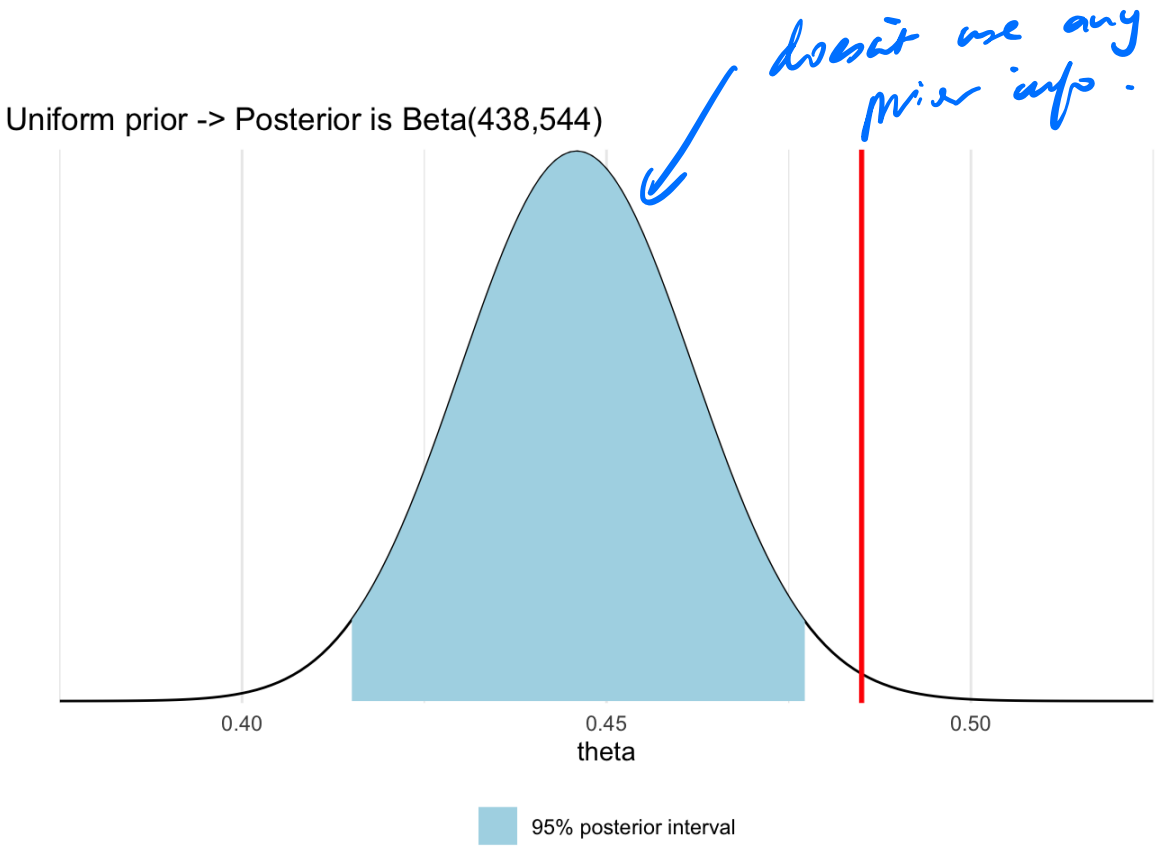
Maternal condition **placenta previa**

- Specific example of a factor that may influence the sex ratio
- Unusual condition of pregnancy in which the placenta is implanted low in the uterus, obstructing the fetus from a normal vaginal delivery.
- Study concerning the sex of placenta previa births in Germany found that of a total of 980 births,

  437 were female, and 543 male,

  so a rate of 0.446 female births.
- How much evidence does this provide for the claim that the proportion of female births in the population of placenta previa births is less than 0.485, the proportion of female births in the general population?

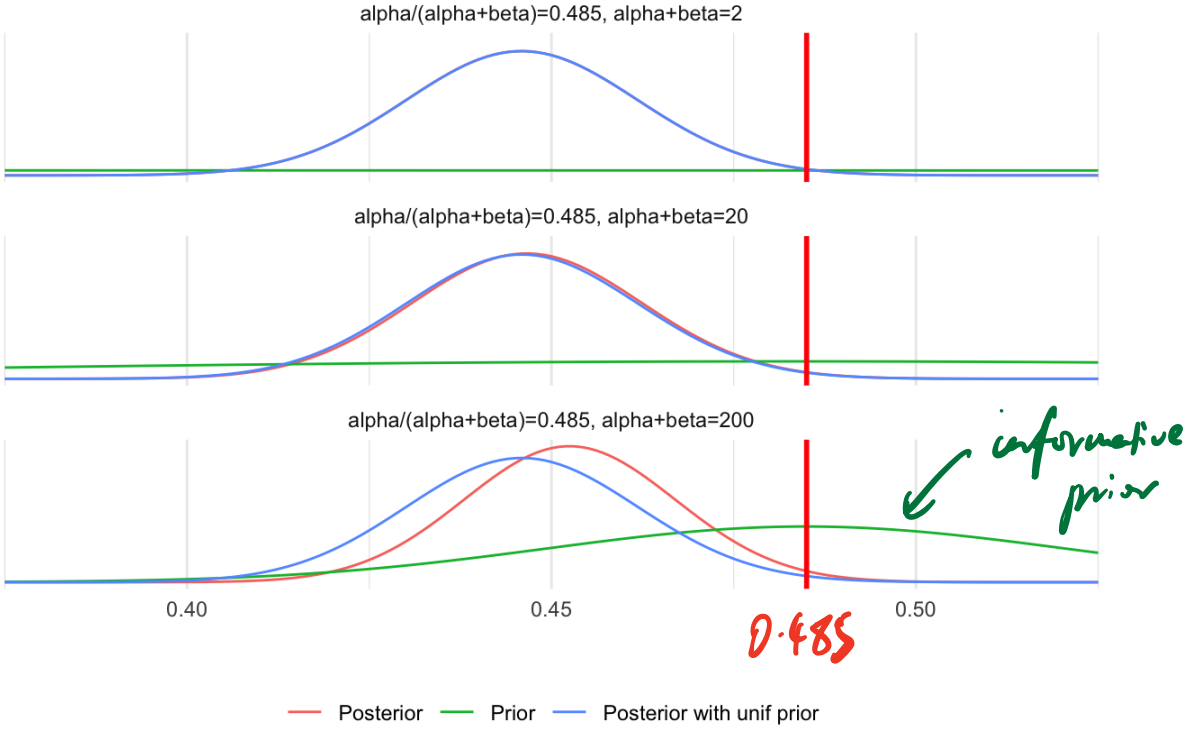
# Example: Beta-Bernoulli–Placenta previa




Adapted from BDA3–Gelman et al. pg 37

# Example: Beta-Bernoulli–Placenta previa

Effect of different priors. $\alpha + \beta$ increasing puts more weight on the prior with mean 0.485. ($\alpha + \beta = 2$ gives a nearly uniform prior.)




Adapted from BDA3–Gelman et al. pg37

# Example: Beta-Bernoulli–Placenta previa

With the uniform prior $P \sim \mathsf{Unif}(0, 1)$, we found

$$P \mid \boldsymbol{X} \sim \mathrm{Beta}(438, 544).$$

The posterior probability answering our original question is

$$\mathbb{P}[P < 0.485 \mid \boldsymbol{X}] = \int_0^{0.485} f_{P|\boldsymbol{X}}(p \mid \boldsymbol{X})\, \mathrm{d}p \approx \boxed{0.993}.$$

**Conclusion:** Given these data and the uniform prior, there is about 99.3% posterior probability that the proportion of female births among placenta-previa births is below 0.485.

The other priors shown give the same qualitative conclusion.

BDA3–Gelman et al., Section 2.4, pp. 37–38.

# Outline

Conjugate priors

Point estimates

More posterior examples

Bayes credible sets

# Definition

A family of priors $f_\Theta(\theta)$ are called **conjugate priors** for a parametric model $f_{\boldsymbol{X}|\Theta}(\boldsymbol{x} \mid \theta)$ if the resulting posteriors $f_{\Theta|\boldsymbol{X}}(\theta \mid \boldsymbol{x})$ have the same parametric form as the priors.

- Beta$(\alpha, \beta)$ are conjugate priors for the model Bernoulli$(p)$.
- Gamma$(\alpha, \beta)$ are conjugate priors for the model Pois$(\lambda)$.
- Normal priors are conjugate for the model $\mathcal{N}(\mu, \sigma^2)$ with unknown $\mu$ and known $\sigma$.
- Gamma$(\alpha, \beta)$ are conjugate priors for the model $\mathcal{N}(\mu, 1/\xi)$ with known $\mu$ and unknown precision $\xi = 1/\sigma^2$.

**Pros**: Conjugate priors exist for many simple parametric models. They are mathematically/computationally convenient because lead to posteriors that are common distributions and easy to compute.

**Cons**: This convenience does not necessarily imply desirable statistical properties. Conjugate priors may limit us to a narrow family that cannot encode prior knowledge like sparsity or multi-modality

# Example: Normal prior for normal mean

Let $X_1, \ldots, X_n \sim \mathcal{N}(\theta, \frac{1}{\xi})$ i.i.d., where for convenience we parametrize the model by the precision $\xi = \frac{1}{\sigma^2}$, instead of the variance $\sigma^2$.
Suppose $\xi$ is known, $\theta$ is unknown with prior $\Theta \sim \mathcal{N}(\mu_0, \frac{1}{\xi_0})$.

Define

$$\xi_{\text{post}} = n\xi + \xi_0, \quad \text{and} \quad \mu_{\text{post}} = \frac{\xi \sum_{i=1}^n X_i + \xi_0 \mu_0}{n\xi + \xi_0}$$

Then the posterior is

$$\Theta | \boldsymbol{X} \sim \mathcal{N}\left(\mu_{\text{post}}, \frac{1}{\xi_{\text{post}}}\right).$$

Derivaton: in homework

# Outline

Conjugate priors

Point estimates

More posterior examples

Bayes credible sets

# Bayesian point estimates

In many applications, we may want a single estimate $\widehat{\theta}$ and an interval that quantifies our uncertainty about $\theta$, instead of a full posterior distribution.

The **posterior mean** and **posterior mode** are the mean and mode of the posterior distribution of $\Theta$, and either may be used as a Bayesian estimate $\widehat{\theta}$.

# Recap: MSE and best estimators

**Best estimators**

Let $X$ and $Y$ be any two random variables.

- We want to use $X$ (observed) to predict $Y$ (unobserved).
- The best estimator, the one minimizing the MSE $\mathbb{E}[(\delta(X) - Y)^2]$, is the conditional expectation

$$\delta(X) = \mathbb{E}[Y \mid X].$$

# Bayes optimality

**Translation to Bayesian setting**

- $\Theta$ is our unknown random parameter
- Data $\boldsymbol{X} \sim f_{\boldsymbol{X}|\Theta}$.

The MSE in this setting is

$$\mathrm{MSE}(\delta) = \mathbb{E}\big[(\Theta - \delta(\boldsymbol{X}))^2\big].$$

This averages the squared error over both the random parameter and the data.

- The best estimator, the one minimizing this MSE over all possible estimators $\delta$, is the **posterior mean**

$$\delta_{\mathrm{Bayes}}(\boldsymbol{X}) = \mathbb{E}[\Theta \mid \boldsymbol{X}].$$

# Example: Beta prior for Bernoulli

Let $X_1, \ldots, X_n \sim \mathsf{Bern}(p)$ i.i.d. with prior $P \sim \mathrm{Beta}(\alpha, \beta)$. We found that the posterior distribution is

$$P \mid \boldsymbol{X} \sim \mathrm{Beta}(S + \alpha, n - S + \beta), \qquad S = X_1 + \cdots + X_n.$$

The posterior mean is

$$\widehat{p} = \mathbb{E}[P \mid \boldsymbol{X}] = \frac{S + \alpha}{n + \alpha + \beta}$$

This is different from the MLE/MoM estimator $\overline{X} = S/n$. It is biased towards the prior mean.

$\hat{p}$ is the sample mean as if observed ($\alpha$ priori) $\alpha$ heads and $\beta$ tails before seeing the data.

# Example: Beta prior for Bernoulli

**Useful interpretation:** $\widehat{p}$ is the weighted average of the sample mean and prior mean.

$$\widehat{p} = \frac{S+\alpha}{n+\alpha+\beta} = \underbrace{\frac{n}{n+\alpha+\beta}}_{\text{sample weight}} \cdot \underbrace{\frac{S}{n}}_{\bar{X}\text{ sample mean}} + \underbrace{\frac{\alpha+\beta}{n+\alpha+\beta}}_{\text{prior weight}} \cdot \underbrace{\frac{\alpha}{\alpha+\beta}}_{\text{prior mean}}$$

For (fixed) prior $\text{Beta}(\alpha,\beta)$, as $n \uparrow$,

sample weight $\nearrow 1$ and prior weight $\searrow 0$

# Outline

Conjugate priors

Point estimates

More posterior examples

Bayes credible sets

# Example: Gamma prior for Poisson

Let $X_1, \ldots, X_n \sim \text{Pois}(\lambda)$ i.i.d. We model $\lambda$ as random with prior $\Lambda \sim \text{Gamma}(\alpha, \beta)$. Derive the posterior.

$$\text{prior}: f_\Lambda(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \quad \text{for } \lambda > 0$$

$$\mathbb{P}[\text{Pois}(\lambda) = k] = \frac{e^{-\lambda}\lambda^k}{k!}$$

$$\text{likelihood of } x_i: \quad f_{X_i|\Lambda}(x_i|\lambda) = \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} \quad (= \mathbb{P}[X_i = x_i])$$

Hence

$$f_{\Lambda|\underline{X}}(\lambda|x_1,\ldots x_n) \propto f_{\underline{X}|\Lambda}(x_1,\ldots,x_n|\lambda) \cdot f_\Lambda(\lambda)$$

$$\propto \left(\prod_{i=1}^n e^{-\lambda}\lambda^{x_i}\right) \cdot \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$\propto \lambda^{s+\alpha-1} e^{-(n+\beta)\lambda} \qquad s = x_1,\ldots,x_n$$

Deduce Posterior $\Lambda|\underline{X} \sim \text{Gamma}(S+\alpha, n+\beta)$

# Example: Gamma prior for Poisson

Let $X_1, \ldots, X_n \sim \text{Pois}(\lambda)$ i.i.d. with prior $\Lambda \sim \text{Gamma}(\alpha, \beta)$. We found that the posterior is

$$\Lambda \mid \boldsymbol{X} \sim \text{Gamma}(S + \alpha, n + \beta), \qquad S = X_1 + \cdots + X_n$$

Write down the posterior mean and interpret it.
(Application: adjusting kidney cancer rates. See BDA3–Gelman et al. pg 47 for details on the adjustment in last lecture's example.)

Posterior mean

$$\hat{\lambda} = \frac{S+\alpha}{n+\beta}$$

as if we had observed ($\alpha$ prior) $\beta$ additional samples with sum $\alpha$

Interpret

$$\hat{\lambda} = \underbrace{\frac{n}{n+\beta}}_{\text{sample weight}} \cdot \underbrace{\frac{S}{n}}_{\bar{X}\text{ sample mean}} + \underbrace{\frac{\beta}{n+\beta}}_{\text{prior weight}} \underbrace{\frac{\alpha}{\beta}}_{\text{prior mean}}$$

# Example: Gamma prior for Normal variance

Let $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \frac{1}{\xi})$ i.i.d., where for convenience we parametrize the model by the precision $\xi = \frac{1}{\sigma^2}$, instead of the variance $\sigma^2$.
Suppose $\mu$ is known, $\xi$ is unknown with prior $\Xi \sim \text{Gamma}(\alpha, \beta)$.

Prior $f_\Xi(\xi) = \frac{\beta^\alpha}{\Gamma(\alpha)} \xi^{\alpha-1} e^{-\beta\xi}, \quad \xi > 0$

likelihood of $X_i$: $f_{X_i|\Xi}(x_i|\xi) = \sqrt{\frac{\xi}{2\pi}}\, e^{-\frac{\xi}{2}(x_i-\mu)^2}$

Hence $f_{\Xi|\underline{X}}(\xi|x_1 \cdots x_n) \propto f_{\underline{X}|\Xi}(x_1 \cdots x_n|\xi) \cdot f_\Xi(\xi)$

$$\propto \left(\prod_{i=1}^n \sqrt{\xi}\, e^{-\frac{\xi}{2}(x_i-\mu)^2}\right) \cdot \xi^{\alpha-1} e^{-\beta\xi}$$

$$\propto \xi^{\alpha+\frac{n}{2}-1} e^{-\xi\left(\frac{1}{2}\sum_{i=1}^n (x_i-\mu)^2 + \beta\right)}$$

Posterior $\Xi|\underline{X} \sim \text{Gamma}\left(\alpha + \frac{n}{2}, \frac{1}{2}\sum_{i=1}^n (x_i-\theta)^2 + \beta\right)$

# Outline

Conjugate priors

Point estimates

More posterior examples

Bayes credible sets

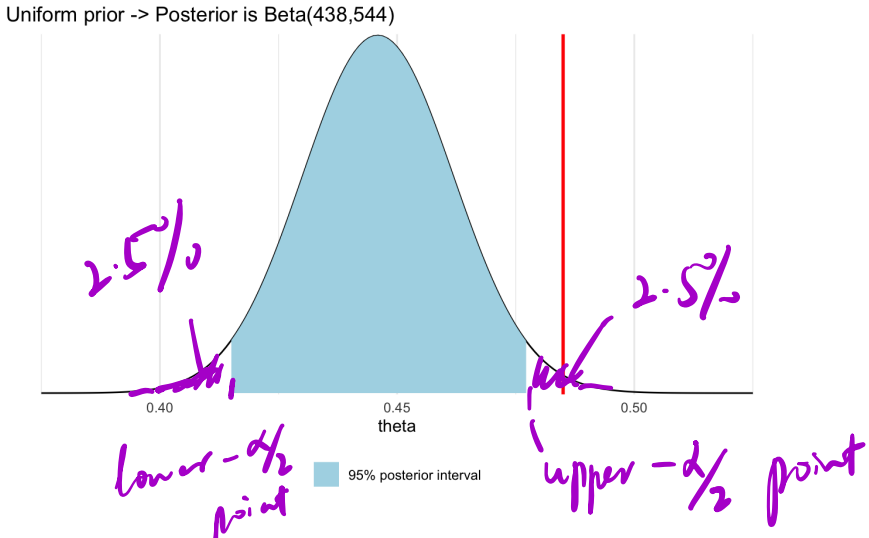
# Bayes credible interval

e.g. 95%

A **Bayesian credible interval** with coverage $1 - \alpha$ is an interval $I(\boldsymbol{X})$ that contains $\Theta$ with posterior probability $1 - \alpha$:

$$\mathbb{P}\left[\Theta \in I(\boldsymbol{X}) \mid \boldsymbol{X} = \boldsymbol{x}\right] = 1 - \alpha,$$

where the probability is over $\Theta$ (instead of over $\boldsymbol{X}$ as in the frequentist confidence interval). Common choice: the interval from the lower-$\alpha/2$ to upper-$\alpha/2$ point of the posterior distribution of $\Theta$.




Adapted from BDA3–Gelman et al. pg 37

# Example: Normal prior for Normal mean

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\theta, 1/\xi)$, where $\xi$ is known and $\theta$ has prior $\Theta \sim \mathcal{N}(\mu_0, 1/\xi_0)$. The posterior will be

$$\Theta \mid \boldsymbol{X} \sim \mathcal{N}(\mu_{\text{post}}, 1/\xi_{\text{post}})$$

where $\xi_{\text{post}} = n\xi + \xi_0$, and the posterior mean is

$$\mu_{\text{post}} = \frac{S + (\xi_0/\xi)\mu_0}{n + (\xi_0/\xi)}, \qquad S = X_1 + \cdots + X_n$$

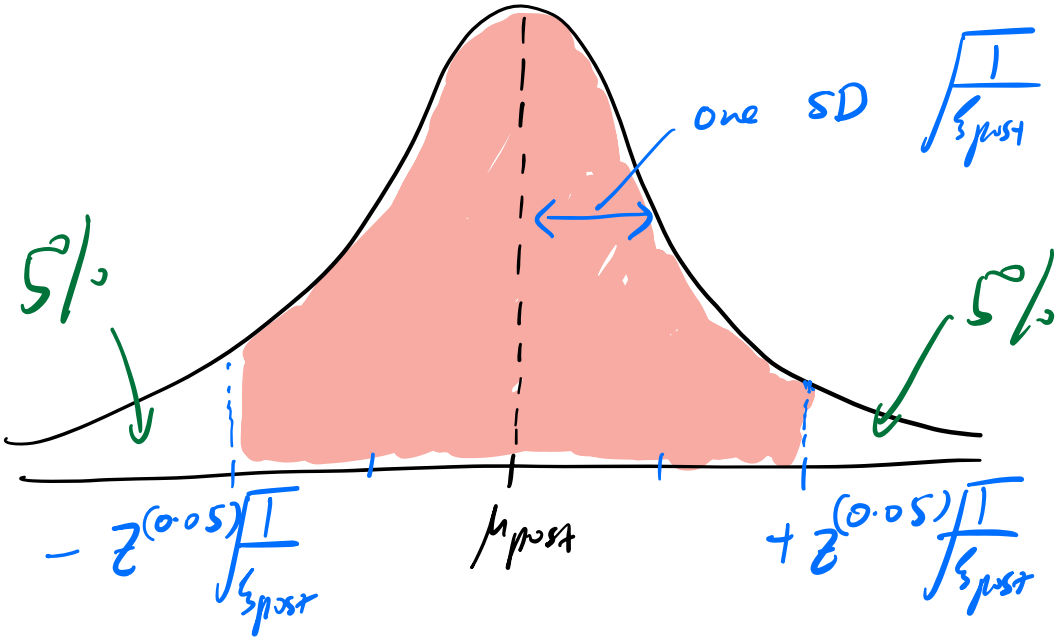
**Interpretation**: it is *as if* we had observed $\xi_0/\xi$ additional samples with mean $\mu_0$, and is again a weighted average

$$\mu_{\text{post}} = \underbrace{\frac{n}{n+(\xi_0/\xi)}}_{\text{sample weight}} \underbrace{\frac{S}{n}}_{\text{sample mean}} + \underbrace{\frac{\xi_0/\xi}{n+(\xi_0/\xi)}}_{\text{prior weight}} \underbrace{\mu_0}_{\text{prior mean}}$$

# Example: Normal prior for Normal mean

The posterior is $\Theta \mid \boldsymbol{X} \sim \mathcal{N}(\mu_{\text{post}}, 1/\xi_{\text{post}})$.

A **90% credible interval for $\theta$** is $\boxed{\mu_{\text{post}} \pm z^{(0.05)} \sqrt{\frac{1}{\xi_{\text{post}}}}}$.



# Credible sets vs confidence intervals

OPTIONAL

A Bayesian credible interval $I(\boldsymbol{X})$ with coverage $1-\alpha$ is an interval which guarantees

$$\mathbb{P}[\Theta \in I(\boldsymbol{X}) \mid \boldsymbol{X} = \boldsymbol{x}] = 1-\alpha.$$

Here $\Theta$ is random, and $1-\alpha$ is the probability of $\Theta \in I(\boldsymbol{x})$ over the randomness of $\Theta$, given the observed data $\boldsymbol{X} = \boldsymbol{x}$.

A frequentist confidence interval $I(\boldsymbol{X})$ with coverage $1-\alpha$ is an interval which guarantees

$$\mathbb{P}_\theta[\theta \in I(\boldsymbol{X})] = 1-\alpha.$$

The parameter $\theta$ is fixed, and $1-\alpha$ is the probability of $I(\boldsymbol{X}) \ni \theta$ over the randomness of $\boldsymbol{X} \sim f(\boldsymbol{x} \mid \theta)$.

# Example: Intervals for Normal location model

OPTIONAL

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$, where the variance is known and equal to 1.

In a frequentist analysis, we might estimate $\theta$ by its MLE $\widehat{\theta} = \overline{X}$. Then $\text{Var}_\theta[\overline{X}] = \frac{1}{n}$, and a frequentist confidence interval for $\theta$ would be

$$\boxed{\overline{X} \pm \frac{1}{\sqrt{n}} \cdot z^{(\alpha/2)}}.$$

This guarantees, for **any fixed value** of the true parameter $\theta$,

$$\mathbb{P}_\theta\left[\, \theta \in \overline{X} \pm \frac{1}{\sqrt{n}} \cdot z^{(\alpha/2)} \,\right] = 1 - \alpha,$$

where $\mathbb{P}_\theta$ is over the randomness of $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$.

# Example: Intervals for Normal location model

In Bayesian inference, suppose we choose the prior distribution

$$\Theta \sim \mathcal{N}(0, 1/\xi_0)\,.$$

OPTIONAL

The posterior distribution is then

$$\Theta \mid \boldsymbol{X} \sim \mathcal{N}\left(\mu_{\text{post}}, \frac{1}{\xi_{\text{post}}}\right),$$

where

$$\mu_{\text{post}} = \frac{n}{n+\xi_0}\,\overline{X}, \qquad \xi_{\text{post}} = n + \xi_0.$$

A Bayesian credible interval would be

$$I_{\text{Bayes}}(\boldsymbol{X}) = \boxed{\frac{n}{n+\xi_0}\,\overline{X} \;\pm\; \frac{1}{\sqrt{n+\xi_0}} \cdot z^{(\alpha/2)}.}$$

- This is centered at the posterior mean, which shrinks $\overline{X}$ towards the prior mean of 0.
- Its width is narrower than the frequentist interval.

# Example: Intervals for Normal location model

OPTIONAL

The Bayes credible interval guarantees that for any realization of the data $\boldsymbol{x} = (x_1, \ldots, x_n)$

$$\mathbb{P}\left[\Theta \in I_{\text{Bayes}}(\boldsymbol{X}) \mid \boldsymbol{X} = \boldsymbol{x}\right] = 1 - \alpha,$$

where $\mathbb{P}$ is over the posterior distribution $\Theta \mid \boldsymbol{X} = \boldsymbol{x}$.

Q: Suppose the true parameter is fixed at $\theta$. Does this $I_{\text{Bayes}}$ cover $\theta$ with probability $1 - \alpha$, in the frequentist sense (over the randomness of the data $\boldsymbol{X}$)? i.e. $\mathbb{P}_\theta\left[I_{\text{Bayes}}(\boldsymbol{X}) \ni \theta\right] = 1-\alpha$ ?

A: Not necessarily. This coverage probability depends on the value of $\theta$. Intuitively if $\theta = 1000$ truly, but our prior mean is 0, this should throw off the frequentist coverage probability.

# Example: Intervals for Normal location model

OPTIONAL

By some manipulation, the coverage probability of the Bayes credible interval $I_{\text{Bayes}}$ in the frequentist sense is

$$\mathbb{P}_\theta\left[\theta \in I_{\text{Bayes}}(\boldsymbol{X})\right] = \mathbb{P}_\theta\left[\theta \in \frac{n}{n+\xi_0}\overline{X} \pm \frac{1}{\sqrt{n+\xi_0}} \cdot z^{(\alpha/2)}\right]$$

$$\vdots$$

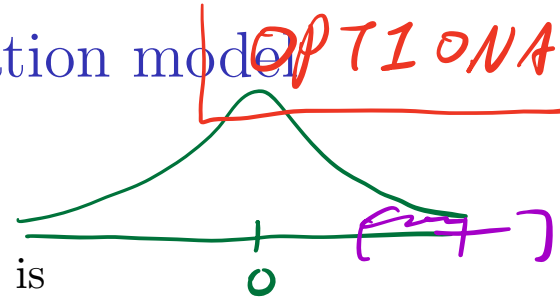
$$= \mathbb{P}_\theta\left[\sqrt{n}(\overline{X}-\theta) \in \frac{\xi_0}{\sqrt{n}}\theta \pm \sqrt{\frac{n+\xi_0}{n}}z^{(\alpha/2)}\right]$$

Now for $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$, we have exactly $\sqrt{n}(\overline{X}-\theta) \sim \mathcal{N}(0,1)$. Thus

$$\mathbb{P}_\theta\left[\theta \in I_{\text{Bayes}}(\boldsymbol{X})\right] = \mathbb{P}_\theta\left[\mathcal{N}(0,1) \in \frac{\xi_0}{\sqrt{n}}\theta \pm \sqrt{\frac{n+\xi_0}{n}}z^{(\alpha/2)}\right]$$

# Example: Intervals for Normal location model

OPTIONAL




The frequentist coverage probability of $I_{\text{Bayes}}$ is

$$\mathbb{P}_\theta\left[\theta \in I_{\text{Bayes}}(\boldsymbol{X})\right] = \mathbb{P}_\theta\left[\mathcal{N}(0,1) \in \frac{\xi_0}{\sqrt{n}}\theta \pm \sqrt{\frac{n+\xi_0}{n}}\, z^{(\alpha/2)}\right],$$

$Z \sim N(0,1)$

- If true parameter $\theta$ is far from 0 (our prior mean), the coverage probability can be smaller than $1-\alpha$.
- On the other hand, if true parameter $\theta = 0$, then the coverage probability is larger than $1-\alpha$ because $\sqrt{(n+\xi_0)/n} > 1$.

So $I_{\text{Bayes}}(\boldsymbol{X})$ has under-coverage when $\theta$ is far from the prior mean, and over-coverage when $\theta$ is close to the prior mean.

# The Bayesian coverage guarantee

OPTIONAL

To summarize, a Bayesian credible interval does not guarantee frequentist coverage for a fixed $\theta$. It instead guarantees coverage in an **average sense**.

Since $\mathbb{P}[\Theta \in I_{\text{Bayes}}(\boldsymbol{X}) \mid \boldsymbol{X} = \boldsymbol{x}] = 1 - \alpha$ for any possible realization of the data $\boldsymbol{X} = \boldsymbol{x}$, the guarantee also holds *unconditionally*:

$$\mathbb{P}[\Theta \in I_{\text{Bayes}}(\boldsymbol{X})] = 1 - \alpha$$

where the probability is over the joint distribution of $(\Theta, \boldsymbol{X})$. The marginal distribution of $\Theta$ is the prior $f_\Theta(\theta)$. We may write

$$\begin{aligned} 1 - \alpha = \mathbb{P}[\Theta \in I_{\text{Bayes}}(\boldsymbol{X})] &= \int \mathbb{P}[\Theta \in I_{\text{Bayes}}(\boldsymbol{X}) \mid \Theta = \theta] \, f_\Theta(\theta) \, \mathrm{d}\theta \\ &= \int \mathbb{P}_\theta[\theta \in I_{\text{Bayes}}(\boldsymbol{X})] \, f_\Theta(\theta) \, \mathrm{d}\theta \end{aligned}$$

frequentist coverage

# The Bayesian coverage guarantee

OPTIONAL

$$1 - \alpha = \int \mathbb{P}_\theta[\theta \in I_{\text{Bayes}}(\boldsymbol{X})]\, f_\Theta(\theta)\, \mathrm{d}\theta$$

The frequentist coverage probability of the Bayes credible interval, when **averaged** over values of $\theta$ according to the prior, is guaranteed to be $1 - \alpha$.

- Suppose we conduct multiple experiments where the parameter $\theta$ is the **same** across all experiments. If we compute 90% frequentist confidence intervals in each experiment, then roughly 90% of these intervals would cover $\theta$. Similar guarantees may not hold for a 90% Bayesian credible interval.
- Suppose instead that $\theta$ is **different** in each experiment, and its distribution across experiments is correctly described by the prior $f_\Theta(\theta)$. If we compute 90% Bayesian credible intervals in each experiment under this prior, then roughly 90% of those intervals would cover the corresponding values of $\theta$.